I. Introduction

dPCR is a widely used molecular quantification technique, with wide applicability to QC procedures especially. It offers significant advantages over quantitative real-time PCR (qPCR).

Real time quantitative PCR (qPCR) relies on comparing the exponential curve of the concentration of an analyte undergoing PCR to a reference, in order to estimate the starting amount of the analyte (usually a nucleic acid). The technique has wide-ranging use-cases in QC contexts; detecting viral RNA or DNA in a sample

However, conditions within the PCR experiment can differ from the conditions present when the reference curve was created, often without the experimenter's knowledge.

dPCR addresses this partitioning the sample across tens of thousands of micro-wells, and performing the PCR procedure in each well individually, the analyte's true concentration in the original sample can thus be discerned.

But dPCR's true innovation is to skip the step of calculating the actual number of molecules within each well. To do so, the experimenter uses the Poisson distribution.

The Poisson distribution asks "given λ, an expected number of occurrences of an event across an interval (in time or space), what is the probability that the event will occur *k* times in that interval"? One of its earliest uses was to model the number of soldiers injured in a given year by kicking horses in the Prussian Army!

A subtlety of the Poisson distribution is that λ can be a decimal – it makes sense to say that 'rain is expected to fall in the desert 0.1 times per year', which means that rain falls approximately once every ten years – while k is always a nonnegative integer; it cannot have rained 0.1 times *in a particular year*.

In the context of dPCR, the partitions are the "intervals" and the "events" are the number of molecules. The average partition should have λ molecules of the analyte in it; the probability that it has *k* molecules can be modelled according to the Poisson distribution. By setting k to 0, an approximate value for the initial concentration of molecules in the sample as a whole can be calculated, by only measuring the number of "empty" partitions – partitions where the target molecule is not present.

Hence, the experimenter does not need to measure anything save presence and absence of the amplicon across the tens or hundreds of thousands of wells. Presence and absence are recorded as binary digits – 1 and 0 – and lend the procedure its name.

However, dPCR's sensitivity depends on the number of partitions and the initial concentration of the target in complex ways. Furthermore, false positives and false negatives further bias the results.

II. Simulation

A simulation was created using Python to model dPCR. Parameters for the simulation included p, the number of partitions; n, the true initial number of molecules of interest in the sample; f_n, the per molecule false negative rate; and f_p, the false positive rate. Furthermore, two 'modes' were created in the simulation; multinomial and Poisson, whose meaning is further elaborated below. A seed for the program's random number generator, in order to ensure reproducibility, constitutes the simulation's sixth and final parameter.

The program works by dividing the true initial number of molecules in the sample by the number of partitions to calculate an expected value for the number of molecules in each partition, referred to as $\lambda = \frac{n}{p}$. Then, this number is used to fill an array of simulated partitions by using either Poisson sampling (if in Poisson mode) or Multinomial sampling (if in Multinomial mode).

In the latter mode, the number of molecules implied by the true initial λ is reproduced exactly; in the former, an approximate estimate is reproduced. The Poisson mode matches laboratory conditions more closely, since taking some sample of a (generally liquid) medium with known concentration will not reproduce the medium's concentration exactly.

The array of partitions is passed through a function that models the reading of binary presence or absence. Within this function, random error is also simulated – both false negatives (which depend on the count of molecules in the partitions;) and false positives (which are fully random). This process is referred to as digitization.

It should be noted that the process of amplification is not itself simulated. This is because amplification is only used in dPCR to identify partitions where the molecule is present – the actual partition-by-partition concentration of the molecule, post-amplification, is immaterial to the process.

From the digitized array of partitions, the rate of empty partitions is calculated. The Poisson distribution is then used with this rate as input (i.e., with k=0 as input) to return an approximate value of λ, referred to as lambda_hat in the code, or $\hat{\lambda}$ in text. If the number of empty partitions is designated p_0, then $\hat{\lambda} = -ln(\frac{p_0}{p})$.

The relative difference between the "true" λ – which is equal to the number of molecules in the initial concentration divided by the number of partitions, and is thus the expected value of molecules per partition – and the estimate $\hat{\lambda}$, will serve as the main subject of experiments using the simulation.

This variable is expected to be sensitive to both the number of partitions and to the true value of λ itself.

For lower values of λ, the sensitivity of experimental error to a fixed false negative rate is expected to be higher, due to the higher false negative rate for partitions with low concentrations of the molecule of interest.

$\hat{\lambda}$ will likely be far less accurate for large values of λ or small numbers of partitions. In the former case, saturation adds noise to the estimate; in the latter, false negatives and positives will have a greater effect on the accuracy of calculations.